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The Impact of Short-Form Video Analytics on Consumer Engagement and Purchase Intention



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1. INTRODUCTION

Social media platforms like TikTok, Instagram Reels, Facebook Reels, and YouTube Shorts have ushered in a new era of communication for businesses and their audiences. With its ability to quickly engage viewers, communicate in a compelling way, and sway consumer decisions, short-form video has emerged as one of the most effective tools in the world of digital marketing. Due to this reason, video marketing campaigns are becoming a popular method for businesses to increase their products' visibility, build brand recognition, and engage with customers, with the focus on shorter videos. [1][2][3][6][7][8]

Short-form videos have made analytics metrics like views, likes, comments, shares, watch time, click-through rate and audience retention even more important for a marketer. These analytics offer insights into user engagement with content and aid businesses in assessing their marketing strategies. These metrics can provide insights into customer preferences, which can help organizations fine-tune their content to better engage their audience and boost marketing results. [5][6][7][8]

But, with the rise of short-form video marketing, figuring out how analytics metrics can drive engagement and purchase intent is still a challenge. There are lots of videos in which the views, likes and comments have been good, but the sales don't follow. It can be difficult for businesses to determine which metrics are the most crucial to track when it comes to impacting consumer behavior and driving purchases. [5][6][7][8]

Thus, it is important to have a better understanding of the correlation between short-form video analytics, consumer engagement, and consumer purchase intent. Understanding those factors behind increased engagement and improved purchase intent can help businesses make better decisions regarding their data-driven marketing strategies, boost their conversion rates, and become more effective at engaging with and converting their customers. The purpose of this research is to examine the effect of short form video analytics on consumer engagement and purchase intention of social media users. [1][5][6][7][8]

2. BACKGROUND OF THE STUDY

2.1 Growth of Short-Form Video Platforms and Digital Marketing

In a digital marketing landscape rapidly evolving with mobile internet technology and social media platforms, short-form video applications like TikTok, Instagram Reels, Facebook Reels, and YouTube Shorts have completely reshaped the marketing environment. These platforms allow users to consume extremely engaging short-duration videos that impact attention, emotions, and decision making. Short videos are becoming a staple of marketing for businesses, as they have become a valuable tool for product promotion, brand visibility and better engagement with the customer. But despite the rise in these platforms, there remains a hurdle in comprehending the relationship between short-form video exposure and consumer engagement and purchase action. [1][2][3][6][7][8]

2.2 Importance of Short-Form Video Analytics in Consumer Behavior

Short-form video platforms produce a ton of user interaction data every day, such as views, likes, comments, shares, watch time, click-through rate, and audience retention. These analytics can help marketers understand consumer video interactions and the effectiveness of their video content. This might suggest good engagement rates, but it doesn't always translate to sales. The research on the effect of these metrics on consumer engagement and purchase intention has thus emerged as an important field of study. [5][6][7][8]

2.3 Influence of Short-Form Video Content on Engagement and Purchase Intention

Past research indicates that short video videos positively affect consumers' engagement and purchase intent. Shen and Wang (2024) revealed that creating emotional connections and persona perceptions between content creators and users has a positive effect on purchase intention. Likewise, Wang and Oh (2023) found that the entertainment value, information quality and social interaction of short-form videos are important factors influencing consumers' sustained purchase intentions. Furthermore, platforms like TikTok have a significant impact on e-commerce consumer behavior, as they drive product awareness and boost purchase intent, as noted by Hoi and Yin (2023). [7][8][3]

2.4 Factors Influencing Consumer Engagement in Short-Form Videos

The psychological and social reasons to consider consumer engagement in short-form video platforms. According to Chen et al. (2023), factors that positively contribute to engagement include the entertainment value, performance expectancy, and social relationships in short-video advertisements. Additionally, influencers' credibility, trustworthiness, social presence, and personalized recommendations can significantly boost user engagement, including likes, comments, shares, and watch time, as per other studies. These engagement factors can have a significant impact on consumer reactions to digital marketing content. [8][4][5]

2.5 Role of Data Analytics in Short-Form Video Marketing

As big data and artificial intelligence (AI) technology continue to grow and evolve, companies are turning to data analytics to gain insights into their customers. Short-form video analytics like engagement rate, watch time, retention rate, and CTR assist marketers in analyzing video performance and refining their marketing strategies. Although there are large scale data available, predicting actual purchase behavior is complex because of psychological, emotional and social influences. Thus, utilizing analytics and behavioral research are crucial pieces to understand consumer decision making process. [5][6][7][8]

2.6 Dataset Used for the Study

The shortform_video_engagement_3000 dataset will be the prime data source for this study. The data shows some 3,000 engagement metrics for short-form video content on platforms like TikTok, Instagram Reels and YouTube Shorts. View counts, likes, comments, shares, watch time, click through rate (CTR), audience retention, engagement score, content category, and the time of the posting. These variables offer a granular view of the way users are interacting with the content and the performance of the video by content type. The data set is realistic engagement data, appropriate for analysis of the relationship between short-form video analytics, consumer engagement and purchase intent. It will be utilized for analyzing the engagement metrics and consumer behaviors through statistical methods to look for patterns, correlations and predictive relationships. [5][6][7]

2.7 Summary

To conclude, the short-form video revolution has revolutionized the digital marketing landscape, bringing in highly engaging video formats with sophisticated analytics systems. Although engagement metrics offer insight into consumer behavior, there is still a lack of understanding of how engagement measures relate directly to purchase intention. To fill this gap, the current study seeks to understand the influence of short-form video analytics on consumer engagement and purchase intention based on the shortform_video_engagement_3000 dataset. [1][5][6][7][8]

3. LITERATURE REVIEW

3.1 Short-Form Video Marketing and Consumer Behavior

The brief video marketing has emerged as a potent device to sway consumer behavior on digital platforms. Research indicates that social media platforms like TikTok, Instagram Reels, and YouTube Shorts greatly influence consumer perceptions of products and brands. According to Lin, short video content boosts the visibility of products and adds power to consumer decisions through visual narration and emotional appeal. Likewise, Gao and Wu (2024) identified that short video ads significantly impact purchase intentions through their ability to deliver engaging and persuasive information to grab users' attention. [6][2]

3.2 Consumer Engagement in Short-Form Video Platforms

To unlock the power of short-form video marketing, it is essential to understand how well the video is engaging the audience. Consumer engagement is one of the major factors for the performance of short-form video marketing. The use and gratifications theory is used to explain engagement, which is based on entertainment, information value and social interaction, as described by Jiang and Chen (2025). Chen et al. (2023) also point out that social presence and perceived enjoyment can greatly boost user engagement, including likes, comments, and shares. These engagement metrics are commonly used to gauge the reaction of viewers to marketing material. [4][8]

3.3 Short-Form Video Content and Purchase Intention

There are many studies that have proven that there is a correlation between short-form video content and intention to buy. Shen and Wang (2024) determined that emotional bonding and values of creators and users have a significant impact on purchasing behavior. Wang and Oh (2023) discovered that product-related short videos have a positive effect on the continuous purchase intention, particularly when the content is informative and entertaining. Moreover, Hoi and Yin (2023) validate that TikTok content is a significant factor in ecommerce buying decisions. [7][8][3]

3.4 Role of Analytics in Short-Form Video Marketing

These metrics – which include views, likes, comments, watch time, and retention rate – are vital in understanding consumer behavior in short-form video. According to Deng et al. (2024), engagement measures capture emotional and rational consumer reactions. Gao and Wu (2024), however, claim that high engagement does not necessarily translate to purchase conversion as other psychological and trust-based factors also come into play when making a purchase decision. Thus, one has to deal with the results of analytics very carefully to learn what is going on with actual consumers. [1][2]

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